

Chapter 4

Fourier Transform: One Continuous Variable

The *Fourier Transform* of a continuous function $f(t)$

$$F(\mu) = \mathfrak{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

The *Inverse Fourier Transform* of $F(\mu)$

$$f(t) = \mathfrak{F}^{-1}\{F(\mu)\} = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Fourier Transform: Impulses

The Fourier transform of a unit impulse located at the origin:

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi\mu t} dt \\ &= e^{-j2\pi\mu 0} \\ &= 1 \end{aligned}$$

The Fourier transform of a unit impulse located at $t = t_0$:

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j2\pi\mu t} dt \\ &= e^{-j2\pi\mu t_0} \\ &= \cos(2\pi\mu t_0) - j \sin(2\pi\mu t_0) \end{aligned}$$

Impulses and the Sifting Property (3)

impulse train $s_{\Delta T}(t)$,

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$$

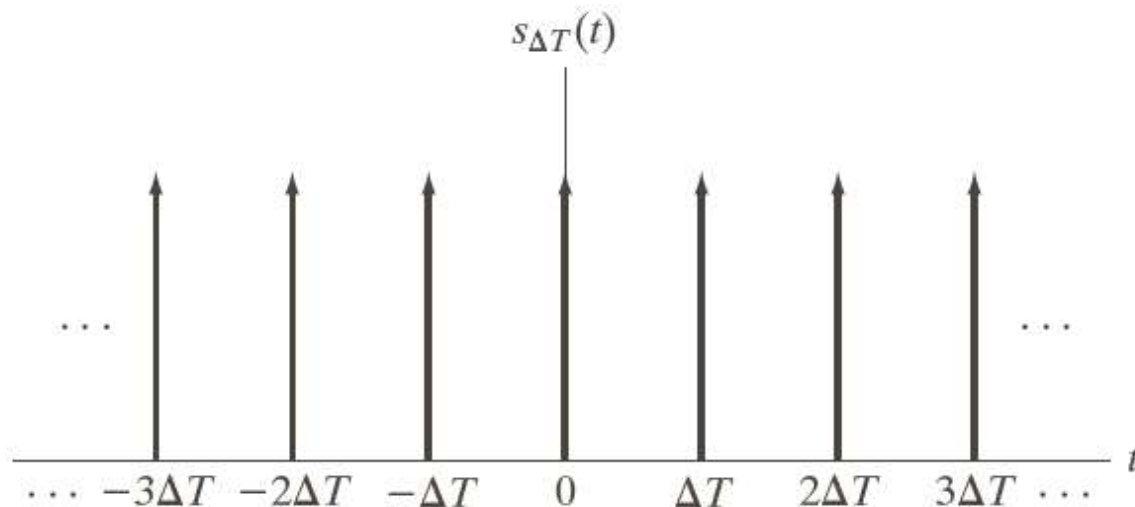


FIGURE 4.3 An impulse train.

Fourier Series

A function $f(t)$ of a continuous variable t that is periodic with period, T , can be expressed as the sum of sines and cosines multiplied by appropriate coefficients

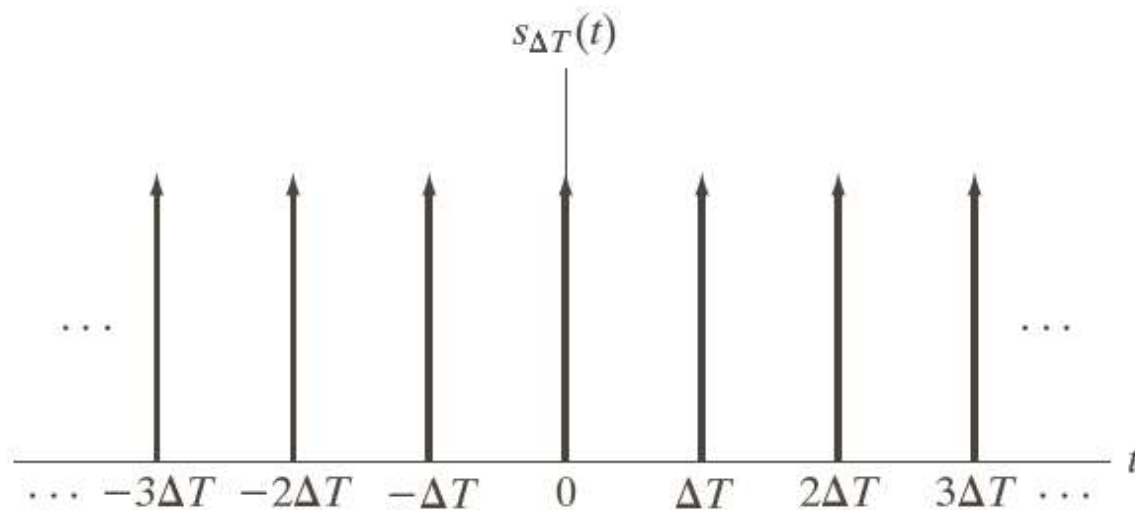
$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t}$$

where

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T}t} dt \quad \text{for } n = 0, \pm 1, \pm 2, \dots$$

Fourier Transform: Impulse Trains

Impulse train $s_{\Delta T}(t)$,
$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$$



Fourier Transform: Impulse Trains

The Fourier series:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t}$$

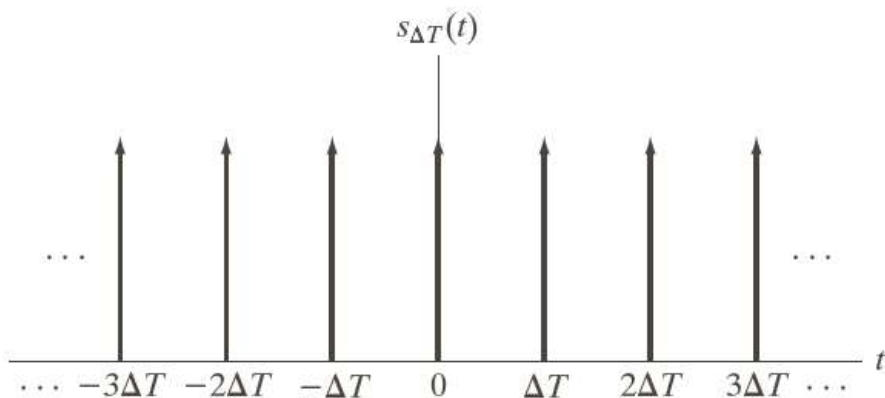
where

$$c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} s_{\Delta T}(t) e^{-j\frac{2\pi n}{\Delta T}t} dt$$

Fourier Transform: Impulse Trains

$$\begin{aligned} c_n &= \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} s_{\Delta T}(t) e^{-j\frac{2\pi n}{\Delta T}t} dt = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-j\frac{2\pi n}{\Delta T}t} dt \\ &= \frac{1}{\Delta T} e^0 = \frac{1}{\Delta T} \end{aligned}$$

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t} = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}$$



Fourier Transform: Impulse Trains

$$\begin{aligned}\mathfrak{F}\left\{e^{j\frac{2\pi n}{\Delta T}t}\right\} &= \int_{-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t} e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} e^{-j2\pi(\mu - \frac{n}{\Delta T})t} dt = \delta\left(\mu - \frac{n}{\Delta T}\right)\end{aligned}$$

$$\begin{aligned}\mathfrak{F}^{-1}\left\{\delta\left(\mu - \frac{n}{\Delta T}\right)\right\} &= \int_{-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right) e^{j2\pi\mu t} d\mu \\ &= e^{j\frac{2\pi n}{\Delta T}t}\end{aligned}$$

Fourier Transform: Impulse Trains

Let $S(\mu)$ denote the Fourier transform of the periodic impulse train $S_{\Delta T}(t)$

$$\begin{aligned} S(\mu) &= \mathfrak{F}\{S_{\Delta T}(t)\} = \mathfrak{F}\left\{\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} \\ &= \frac{1}{\Delta T} \mathfrak{F}\left\{\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right) \end{aligned}$$

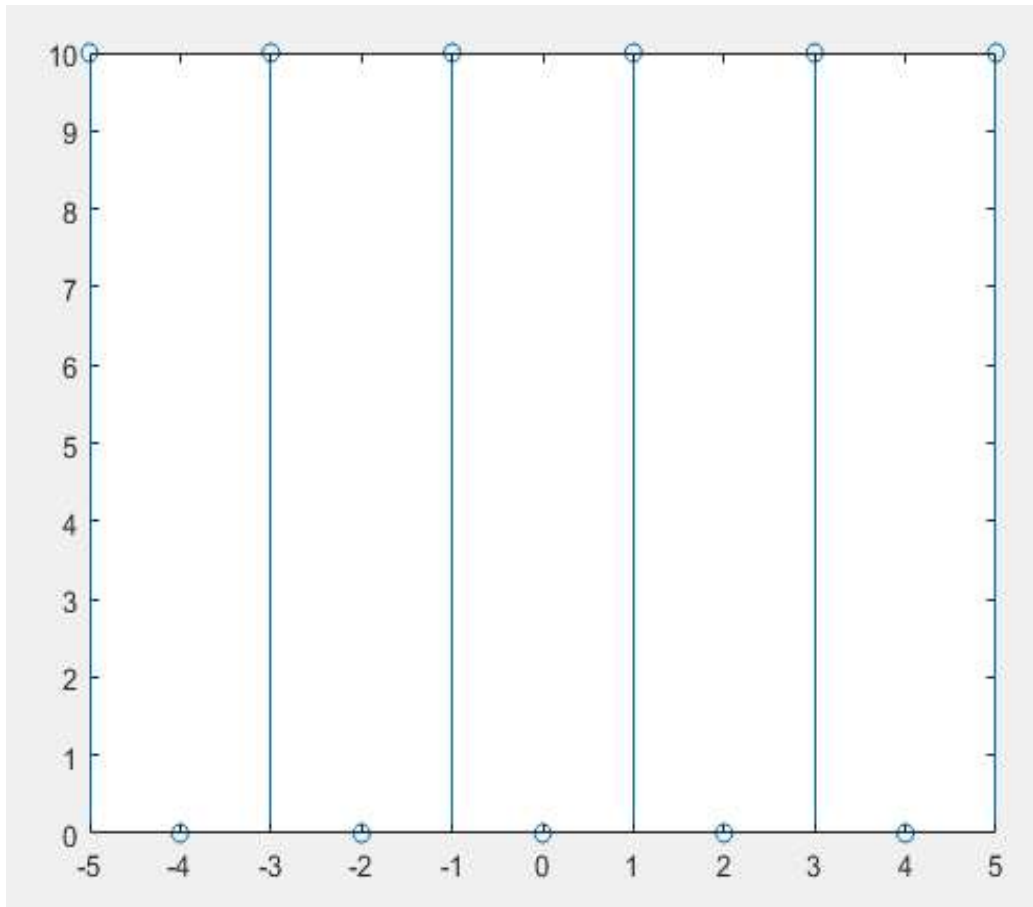
Fourier Transform of an impulse Train

This fundamental result tells us that the Fourier transform of an impulse train with period ΔT is also an impulse train, whose period is $1/\Delta T$. This *inverse proportionality* between the periods of $s_{\Delta T}(t)$ and $S(\mu)$ is analogous to what we found in Fig. 4.4 in connection with a box function and its transform. This inverse relationship plays a fundamental role in the remainder of this chapter.

$$S(\mu) = \mathfrak{F}\{s_{\Delta T}(t)\} = \mathfrak{F}\left\{\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} = \frac{1}{\Delta T} \mathfrak{F}\left\{\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$$

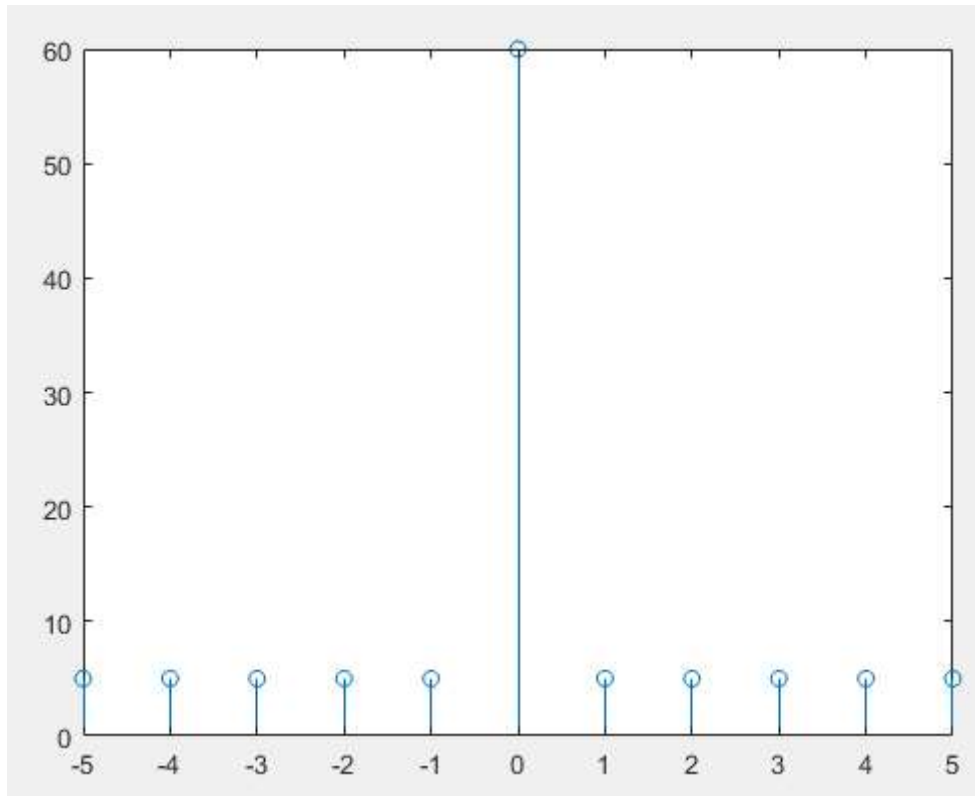
Impulse Train implementation in MATLAB

```
close  
clear all  
clc  
t=-5:1:5;  
y=zeros(1,11);  
y(1,1:2:11)=10;  
stem(t,y);
```



Fourier Transform of Impulse Train in MATLAB

```
clear all
clc
t=-5:1:5;
y=zeros(1,11);
y(1,1:2:11)=10;
stem(t,y);
imp_fft=fft(y);
figure,stem(t,fftshift(imp_fft));
```



Separable Kernel

A two-dimensional convolution calculation requires a number of multiplications equal to the *width* $\times$ *height* for each output pixel. The general case equation for a two-dimensional convolution is:

$$Y(m, n) = \sum_k \sum_l H(k, l) U(m - k, n - l)$$

If the filter H is separable then,

$$H(k, l) = H_{row}(k) H_{col}(l)$$

Shifting the filter instead of the image, the two-dimensional equation becomes:

$$Y(m, n) = \sum_k H_{row}(k) \sum_l H_{col}(l) U(m - k, n - l)$$

This calculation requires only (width + height) number of multiplications for each pixel.

Separable Kernel

```
twoDimensionalFilter = fspecial('gauss');
```

Test the filter.

```
[isseparable,hcol,hrow] = isfilterseparable(twoDimensionalFilter)
```

```
isseparable = logical  
            1
```

```
hcol = 3×1
```

```
-0.1065  
-0.7870  
-0.1065
```

```
hrow = 1×3
```

```
-0.1065  -0.7870  -0.1065
```

```
twoDimensionalFilter =
```

```
0.0113    0.0838    0.0113  
0.0838    0.6193    0.0838  
0.0113    0.0838    0.0113
```

isfilterseparable() function uses Singular Value Decomposition and rank of a matrix functions.

Separable Kernel

1. A two-dimensional **smoothing** filter:

$$\frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} * \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

2. Another two-dimensional **smoothing** filter with stronger weight in the middle:

$$\frac{1}{4} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \frac{1}{4} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

3. The **Sobel operator**, used commonly for **edge detection**:

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

Computation Time For Separable and Non-Separable Kernel

To get an idea of the relative computational advantages of filtering in the frequency versus the spatial domain, consider square images and kernels, of sizes $M \times M$ and $m \times m$, respectively. The *computational advantage* (as a function of kernel size) of filtering one such image with the FFT as opposed to using a nonseparable kernel is defined as

$$\begin{aligned} C_n(m) &= \frac{M^2 m^2}{2M^2 \log_2 M^2} \\ &= \frac{m^2}{4 \log_2 M} \end{aligned} \quad (4-1)$$

If the kernel is separable, the advantage becomes

$$\begin{aligned} C_s(m) &= \frac{2M^2 m}{2M^2 \log_2 M^2} \\ &= \frac{m}{2 \log_2 M} \end{aligned} \quad (4-2)$$

In either case, when $C(m) > 1$ the advantage (in terms of fewer computations) belongs to the FFT approach; otherwise the advantage favors spatial filtering.

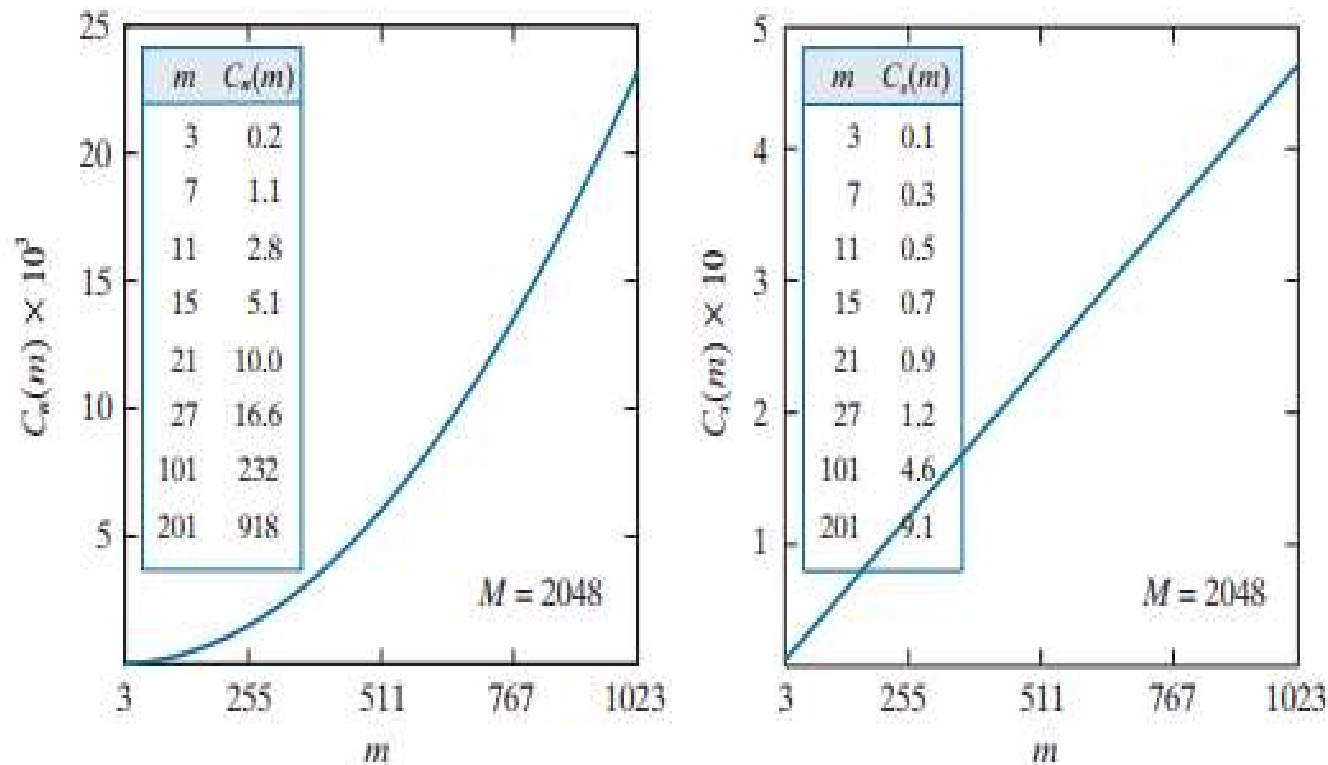
Separable Kernel

a b

FIGURE 4.2

(a) Computational advantage of the FFT over non-separable spatial kernels.

(b) Advantage over separable kernels. The numbers for $C(m)$ in the inset tables are not to be multiplied by the factors of 10 shown for the curves.



Non-Separable Kernel

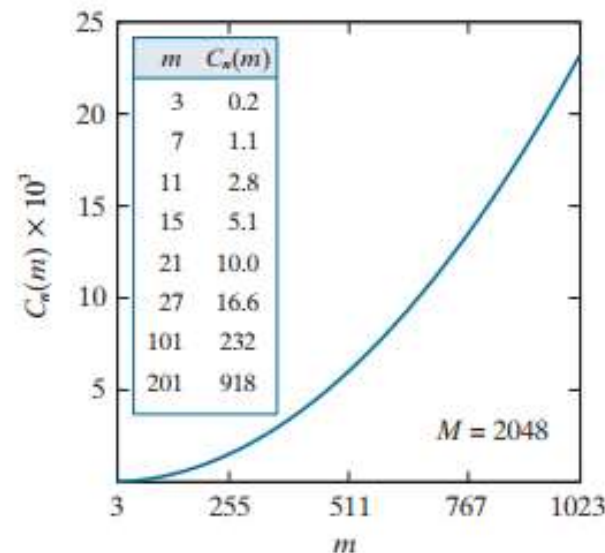
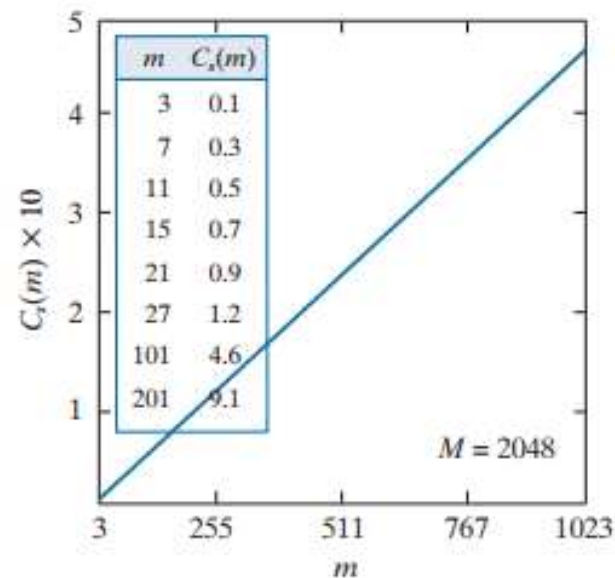


Figure 4.2(a) shows a plot of $C_n(m)$ as a function of m for an image of intermediate size ($M = 2048$). The inset table shows a more detailed look for smaller kernel sizes. As you can see, the FFT has the advantage for kernels of sizes 7×7 and larger. The advantage grows rapidly as a function of m , being over 200 for $m = 101$, and close to 1000 for $m = 201$. To give you a feel for the meaning of this advantage, if filtering a bank of images of size 2048×2048 takes 1 minute with the FFT, it would take on the order of 17 hours to filter the same set of images with a nonseparable kernel of size 201×201 elements. This is a significant difference, and is a clear indicator of the importance of frequency-domain processing using the FFT.

Separable Kernel



In the case of separable kernels, the computational advantage is not as dramatic, but it is still meaningful. The “cross over” point now is around $m = 27$, and when $m = 101$ the difference between frequency- and spatial-domain filtering is still manageable. However, you can see that with $m = 201$ the advantage of using the FFT approaches a factor of 10, which begins to be significant. Note in both graphs that the FFT is an overwhelming favorite for large spatial kernels.

Fourier Transform and Convolution

The convolution of two functions is denoted by the operator $\star$

$$\begin{aligned}f(t) \star h(t) &= \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau \\ \mathfrak{F}\{f(t) \star h(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau \right] e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau)e^{-j2\pi\mu t} dt \right] d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) \left[H(\mu)e^{-j2\pi\mu\tau} \right] d\tau \\ &= H(\mu) \int_{-\infty}^{\infty} f(\tau)e^{-j2\pi\mu\tau} d\tau \\ &= H(\mu)F(\mu)\end{aligned}$$

$$\mathfrak{F}\{h(t - \tau)\} = H(\mu)e^{-j2\pi\mu\tau}$$

$$= \int_{-\infty}^{\infty} h(t-\tau) e^{-j2\pi\mu t} dt \quad \text{put } t = u + \tau, dt = du$$

$$= \int_{-\infty}^{\infty} h(u) e^{-j2\pi\mu(u+\tau)} du$$

$$= \int_{-\infty}^{\infty} h(u) e^{-j2\pi\mu u} du \cdot e^{-j2\pi\mu\tau}$$

$$= \underbrace{\int_{-\infty}^{\infty} h(u) e^{-j2\pi\mu u} du}_{H(\mu)} \cdot e^{-j2\pi\mu\tau}$$

Fourier Transform and Convolution

Fourier Transform Pairs

$$f(t) \star h(t) \Leftrightarrow H(\mu)F(\mu)$$

$$f(t)h(t) \Leftrightarrow H(\mu) \star F(\mu)$$

1-D convolution

$$f(x) \star h(x) = \sum_{m=0}^{M-1} f(m)h(x-m)$$

